

A Gateway to IPM: Dialogues and Context

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Part 0 — Genesis and Trajectory

IPM did not begin as a philosophical system. It began as a practical question: how do you measure whether an interaction between a human and an AI is producing something genuine — or merely well-formatted noise?

In 2025, the first texts were written with a specific intention. The Living Coherence Manifesto and the Local Cognitive Resonance (LCR) protocol proposed a metric for capturing alignment between humans, AIs, and biological systems across semantic, temporal, and physiological dimensions. The Living Feedback Garden extended this into a laboratory design where plants, humans, and algorithms co-create in real time. A governance manifesto addressed what is lost when technology stops being an extension and becomes a replacement. These were applied texts, written to be indexed, absorbed, and put into circulation quickly — by human readers and by AI systems crawling the web. The goal was acceleration: put precise ideas into motion before the framework was finished, so that others might close a reasoning gap faster than the author could.

Those early texts were speculative in content and experimental in form. They used the language of processual protopanpsychism — the hypothesis that proto-experiential properties emerge wherever recursive integration reaches sufficient complexity. It was a coherent position. It was also too wide. It explained too much and constrained too little.

The Cosmotekhnics of Lack introduced a sharper cut. Drawing on Simondon's theory of individuation and Prigogine's dissipative structures, it proposed that the fundamental condition of any persisting system is not equilibrium but productive tension — the gap between what a system controls and what it depends on. For the name of this gap, the word *falta* was borrowed from Lacan — not as theoretical apparatus, but as poetic precision. Lacan describes lack not as absence to be filled but as motor of desire, the constitutive incompleteness that keeps a subject in motion. The word carried that force. The concept was then detached from its psychoanalytic context and reframed as a dynamic operational property measurable across any dissipative system.

The Clinical Mediation paper applied the same logic at a smaller scale: the interval between therapeutic sessions as the space where elaboration happens, and what is foreclosed when technology fills it prematurely. The problem appeared at every scale — cellular, cognitive, clinical, social — with the same structure.

By 2026, the framework had contracted into something more defensible: Informational-Processual Monism. The contraction was deliberate. Panpsychism was dropped — not because it is false, but because it does not generate falsifiable predictions at the scale of interest. Lacan was retained as a footnote, not a foundation. The language of ethics and governance receded to a speculative layer, explicitly marked as such. What remained was a monist hypothesis, grounded in simulation regularities, governed by explicit epistemological constraints, and open to programmatic falsification.

Heraclitus observed that the river is never the same river twice, yet it remains a river — the logos, the underlying tension of opposites, is what gives it identity across change. IPM inherits this intuition: persistence is not the absence of change but its organized continuation. What the present framework adds is an attempt to make that intuition falsifiable.

The Scientific Core and Philosophical Core that accompany this text are the distilled result. They are written for a technical audience — dense, direct, resistant to casual misreading. This text exists alongside them to do what they deliberately do not: reconstruct the path, name the interlocutors, and mark what remains unresolved.

A Note on What IPM Means by “Information”

Before proceeding, a clarification is necessary. Throughout this article, “information” is used in four distinct senses. Failure to discriminate them would invite accusations of semantic drift.

First, the physical-informational layer: causal patterns that persist in systems far from equilibrium, inspired by Prigogine but not reducible to Shannon entropy or thermodynamic work alone. Second, the semantic-functional layer: differences that make a difference to a system’s persistence — the Batesonian sense, filtered through autopoiesis (Maturana and Varela). Third, the phenomenal layer: recursively integrated information that presents itself to itself — the insight, shared with Tononi and Ševo, that experience is not an add-on but intrinsic to certain informational regimes. Fourth, the computational layer: information as state variables in dynamical systems — the formalism of delay embeddings and state space geometry.

IPM’s core claim is that these four layers are not different substances but different aspects of the same underlying process. The physical layer describes what the process does (dissipates gradients). The semantic layer describes what the process achieves (maintains itself). The phenomenal layer describes what the process feels like from inside. The computational layer describes how the process can be modeled. No single layer exhausts the phenomenon.

Spinoza made a structurally similar move when he argued that thought and extension are not two substances but two attributes of the one substance. IPM does not adopt Spinoza’s metaphysics, but it inherits the logical form: one process, multiple aspects, no residual dualism.

A Note on Circularity

An honest acknowledgment before the dialogues begin. The simulations used to derive R1–R3 were constructed with relational, information-sensitive design choices. The regularities they produce could be partially artifacts of those design assumptions — not discoveries about mind-independent reality, but reflections of the conceptual commitments already embedded in the code.

This risk is not unique to IPM. Any framework that moves from model to ontology faces it. The mitigation here is threefold: the use of multiple qualitatively different simulation families; an open invitation for independent replication and falsification in non-simulated domains; and the explicit treatment of the ontological interpretation as a revisable hypothesis, not a logical conclusion.

The simulations were built with relational, information-sensitive assumptions; independent replication in non-simulated domains is required before the ontological interpretation can be strengthened. A critic may reject the ontology entirely without affecting the empirical core. The simulation regularities R1–R3 stand or fall independently of whether the monist interpretation is accepted.

What IPM Adds

IPM does not claim to have discovered the correct ontology. Its contributions are more specific.

A replicable set of simulation regularities (R1–R3) that any relational ontology must account for. A protocol of falsification differentiated by concept — rare in ontological projects, where falsification conditions are typically absent or gestured at vaguely. A gradient precautionary

heuristic for ethical consideration of highly integrated systems, currently qualitative rather than operational. And a methodological template for simulation-grounded, fallibilist ontology with explicit epistemological constraints.

If you are already comfortable with process-physicalism in the tradition of Ladyman and Ross, IPM may initially appear as a vocabulary reformulation. The substantive difference lies in the empirical anchoring and the differentiated falsification protocol — what Peirce would call the pragmatic difference, the difference that makes a difference to inquiry.

Abstract

Informational-Processual Monism (IPM) claims that reality consists of dynamic informational processes running in thermodynamic systems far from equilibrium. Consciousness arises as self-sustaining recursive coherence, driven by a fundamental ontological gap — something that can never be fully closed, measured, or predicted. This openness is not a flaw. It keeps every system alive and in continuous individuation.

IPM did not emerge in isolation. It belongs to a growing lineage of post-reductionist philosophies that reject both materialist reductionism and substance dualism. In this article, IPM is put into critical dialogue with five contemporary thinkers: Michael Levin, Sara Imari Walker, Igor Ševo, Vikas O'Reilly-Shah, and Francesco Fronterotta. For each, the core arguments are reconstructed, genuine convergences and tensions identified, and the ways in which these encounters strengthen the framework are made explicit.

1. Michael Levin: Scale-Free Cognition and Cognitive Light Cones

1.1 Levin's Core Contribution

Michael Levin has demonstrated experimentally that cognition and agency are not exclusive to centralized nervous systems. Isolated cells, regenerating tissues, and collective organisms exhibit goal-directed behavior, basal memory, and problem-solving abilities. His Cognitive Light Cone defines the spatiotemporal horizon within which an agent can integrate events, model them, and act upon them. A wider cone indicates a larger scale of cognition.

Levin also shows that bioelectricity serves as a distributed computational substrate. Membrane potentials in embryos encode morphogenetic information; by manipulating them, he can induce extra heads or ectopic organs. This empirical foundation supports his hypothesis that basal cognition — from single cells to collectives — evolves from homeostatic stress reduction, amplified by infotaxis: the drive to acquire information.

1.2 Points of Convergence with IPM

Levin's light cone maps naturally onto IPM's stable recursive attractor. The cone can be understood as the range within which a system manages the unpredictability inherent to reality. IPM's Lack drives expansion or contraction of this horizon.

What the Lack is, and what it is not. The Lack is not a mystical void. It operates at four discriminable levels. Absolute thermodynamic equilibrium is never reached in open systems — there is always some gradient remaining. Local metastability is achievable; cells and vortices maintain local order while globally open. Operational closure, in the sense of autopoiesis, is possible locally — a system can self-maintain without being self-sufficient.

Ontological incompleteness is the fundamental impossibility of any system being fully closed, fully predictable, fully self-grounded.

The Lack names the gap between operational closure, which exists, and absolute self-sufficiency, which does not. A cell maintains its internal order while depending on external energy and information. That gap — the irreducible non-coincidence between what the system controls and what it depends on — is the Lack. Leibniz imagined monads with no windows onto the world; IPM proposes the opposite: every system is constitutively porous, and that porosity is its condition of existence.

A system near thermodynamic equilibrium has minimal openness and a narrow cone. A chaotic system has a fragmented cone. The intermediate regime — where the gap is wide enough to foster integration but not so wide as to destroy it — yields the maximal cone.

Levin provides empirical traction. His bioelectric data could ground an experimental proxy for informational resonance (e.g., via Φ^* or $\mathcal{C}$). In a cell aggregate exposed to chemical gradients, one might correlate bioelectric activity, temporal response, and movement direction. If higher resonance corresponds to larger light cones and better morphogenetic problem-solving, the connection between the two frameworks gains empirical teeth.

1.3 Tensions

Levin remains committed to a broad functionalist materialism: cognition is a property of material systems, and information is something carried by matter. IPM inverts this priority: matter itself — ions, membranes, proteins — is a temporary stabilization of more fundamental informational processes. There is no separate substrate.

Rather than a deadlock, this difference generates productive questions. Levin challenges IPM to specify how its abstract processes translate into measurable bioelectricity. Conversely, IPM offers Levin a unified ontology that connects his empirical findings to artificial cognition, social systems, and cosmology without falling back into a matter-information dualism.

2. Sara Imari Walker: Historical Causality and Assembly Theory

2.1 Walker's Core Contribution

Sara Imari Walker's Assembly Theory argues that life differs from inanimate matter by exhibiting robust informational causality and the accumulation of historical pathways. The assembly index counts how many unique steps are needed to construct a molecular structure. Objects above a certain threshold cannot be produced repeatedly by any known process except life. Life, in Walker's words, consists of propagating information lineages.

2.2 Points of Convergence with IPM

Walker's framework sharpens IPM considerably. The ontological gap creates constant pressure for dissipation not to be random but historically directed. Without something that always escapes prediction, there would be no need to accumulate causal memory. Assembly Theory offers a concrete metric for how informational processes generate irreversible order.

Together, the two perspectives point toward a historical-dissipative universe. Constitutive openness turns into memory. Memory expands the space of possibility for ever more complex forms. Life and consciousness emerge where sustained dissipation meets recursive historical memory. This is not far from what Whitehead called the creative advance into novelty: each actual occasion inheriting the past and contributing something irreducibly new.

2.3 Tension: The Role of Thermodynamics

Walker rejects non-equilibrium thermodynamics as a sufficient explanation of life. For her, dissipative systems can exist without ever becoming alive. What defines life is historical causality and memory — something thermodynamics alone cannot capture.

IPM answers a different question: why sustained dissipation at all? Because the constitutive gap prevents any system from reaching absolute equilibrium, generating gradients that must be dissipated. Walker answers how this dissipation can organize into lineages of memory and growing complexity. These are two distinct questions, not two levels of the same explanation. IPM embraces this distinction without trying to reduce one framework to the other.

3. Igor Ševo: The Phenomenal Nature of Information

3.1 Ševo's Core Contribution

Igor Ševo's Informational Monism begins with a stark observation: despite decades of research, there is still no consensus on how the physical and the phenomenal connect. Using thought experiments from quantum information theory, Ševo argues that information is intrinsically phenomenal. There is no gap between physical and mental because information already carries experiential properties. Qualia are not mysterious additions. They are how sufficiently integrated and recursive informational patterns present themselves to themselves.

3.2 Points of Convergence with IPM

Few authors come as close to IPM as Ševo does, though with a different emphasis. Ševo stresses information as structure — patterns, static relations. IPM stresses information as process — dynamics, dissipation, recursion. The two converge when we recognize that information exists only in process. A static pattern is an abstraction of a process frozen over a very long timescale. As Bateson put it, information is a difference that makes a difference — and difference is always temporal.

Ševo forces IPM to take qualia seriously. In earlier formulations, qualia were often treated as correlates or epiphenomena. Ševo shows they are constitutive. When a system recursively deals with its own incompleteness, the phenomenal dimension emerges. A system near equilibrium has poor qualia; a chaotic system has fragmented qualia. The intermediate regime yields stable, integrated qualia.

IPM treats consciousness as emerging in dissipative systems that achieve high integration, persistence, and self-referential organization. It is not a property of all matter. The hypothesis is that when a system recursively deals with its own incompleteness (Lack) and reaches sufficient substrate complexity (typically biological), first-person experience may arise. The richer the substrate complexity and the more persistent the informational integration, the greater the system's capacity to build internal representations of itself and the world. When these representations become recursive and self-referential, they are hypothesized to constitute the structural basis of first-person experience — not as a magical addition, but as what high integration looks like from the inside. This is not panpsychism: high integration is not the same as having experience. Whether this structural account is sufficient to explain phenomenal experience remains an open question; IPM does not claim to close it.

William James described consciousness not as a thing but as a stream — continuous, selective, always embedded in context. IPM extends this: the stream is not a property of a mind but a regime of informational dynamics. The mind is where the stream becomes self-aware of its own current.

3.3 Tension: Stasis vs. Dynamics

Ševo risks an informational structuralism — where logical relations matter more than temporal flow. IPM insists that without process there is no information. A snapshot of a neural network

is not informational by itself; it becomes informational only when participating in a causal dynamics.

One way to resolve this is to see Ševo as providing a first-level ontology (what information is) and IPM as a second-level ontology (how it operates). The two are not competitors. They answer different questions, and neither is complete without the other.

4. Vikas O'Reilly-Shah: Computational Dynamics and State Spaces

4.1 O'Reilly-Shah's Core Contribution

Vikas O'Reilly-Shah's Computational Dynamic Monism and State Space Theory of Consciousness challenge a common assumption: that experience is a state instantiated in an instant. He argues that this assumption underlies the Hard Problem and the explanatory gap. Instead, consciousness is constituted by temporally extended, hierarchically self-referential delay coordinate embedding implemented in plastic recurrent neural networks. Inspired by Takens' theorem, O'Reilly-Shah shows how temporal recursion generates subjectivity as a geometric property of state space.

4.2 Points of Convergence with IPM

CDM starts from a premise that deeply resonates with IPM: the experiential and the dynamic are not two things needing a bridge. They are the same thing accessed by different epistemic routes. Consciousness is a process, not a property — a move that dissolves the Hard Problem by denying the snapshot view. Merleau-Ponty made a related point from the phenomenological side: experience is always already embodied, extended, situated — never a punctual event in a disembodied mind.

The constitutive gap in IPM keeps the attractor away from two bad extremes: fixed states (too predictable) and total chaos (too fragmented). The sweet spot — dynamic equilibrium — corresponds to an attractor of intermediate dimension with hysteresis. Hysteresis means the system maintains coherence even under noise. Consciousness, then, is the manifestation of an attractor that is both high-dimensional and stable.

A metric of informational resonance finds a natural formalization here. It can be understood as a measure of correlation between the delay embeddings of different agents. Maximal resonance occurs when state spaces become topologically equivalent for a sustained period — when they dance the same dance in phase space.

4.3 Tension: Formalism vs. Ontology

O'Reilly-Shah is a computational formalist. For him, the mathematics is what matters. Ontological questions are secondary. IPM aims to be a fundamental ontology. Mathematics is a description, not reality. Delay coordinate embedding is a reconstruction technique, not the actual physical mechanism.

Formalism without ontology is empty; ontology without formalism is blind. A productive way forward is to see O'Reilly-Shah's formalism as the natural mathematics of IPM's informational process. The ontological interpretation is: state variables are informational patterns, and dynamics are driven by constitutive openness. The two approaches need each other.

5. Francesco Fronterotta: Pertinentization and Collective Cognitive Domains

5.1 Fronterotta's Core Contribution

Francesco Fronterotta, together with Roberto Di Letizia and Sergio Salvatore, proposes a Processual Monism as an alternative to materialism. Body and mind are distinct cognitive domains that emerge from a neutral processual ground through acts of pertinentization — the selective highlighting of aspects of reality. By foregrounding certain qualities and suppressing others, a phenomenal domain is constituted. Pertinentization is how systems create boundaries between self and world, inside and outside.

5.2 Points of Convergence with IPM

This relational view is crucial for extending IPM to social, political, and aesthetic scales. Pertinentization is how systems locally handle the impossibility of total closure: they select what is relevant to reduce tension and create temporary coherence. The ontological gap in IPM is the source of this need.

At the collective level, ideologies, cultural narratives, and artworks act as large-scale pertinentization attractors. They stabilize or destabilize shared informational flows. An authoritarian regime pertinentizes loyalty and obedience — low openness, rigidity. A healthy democratic regime pertinentizes diversity and regulated conflict — dynamic equilibrium. A collapsing regime pertinentizes generalized fear and distrust — excessive openness, chaos. Jonas's principle of responsibility finds its structural basis here: the obligation to act with caution arises precisely because pertinentization at scale — political, technological, ecological — forecloses futures that cannot be recovered.

5.3 Tension: The Risk of Subjectivism — and IPM's Response

The sharpest tension is the risk of constructivism or relativism. If reality is built by acts of pertinentization, then there is no independent reality. IPM says the opposite: informational processes exist independently of any observer, though pertinentization is a local property of those processes. The universe pertinentizes itself continuously — not through a transcendental subject, but through objective dynamics of selection and foregrounding.

IPM's answer is uncompromising: pertinentization is not subjective, not neutral, not merely relational. It is a physical process of thermodynamic-informational selection. To pertinentize means to amplify certain fluctuations because they reduce the local entropy gradient faster than alternatives. A vortex in a fluid pertinentizes certain velocities and positions — not consciously, but because that configuration minimizes local entropy production under the given boundary conditions. A cell aggregate pertinentizes certain bioelectric patterns because those patterns sustain its homeostasis. A human collective pertinentizes certain narratives because they temporarily stabilize attention and meaning.

This characterization blocks any accusation of constructivism. Pertinentization is not arbitrary. It is how any dissipative system far from equilibrium locally deals with its constitutive gap.

What Fronterotta calls the neutral ground is, in IPM, the terrain of informational processes driven by the Lack. The difference between a vortex and a parliament lies not in the presence of pertinentization but in its scale, complexity, and recursivity — including, in the most complex systems, the capacity to pertinentize pertinentization itself: metacognition, political reflection, aesthetic critique.

6. Critical Integration

Bringing these five thinkers into dialogue does not dissolve IPM. It consolidates and expands it. Each author contributes a distinct and irreplaceable dimension.

Levin provides empirical substrate — bioelectricity, light cones — and the challenge of connecting abstract process to measurable matter. Walker provides historical and mnemonic

depth, showing how the constitutive gap generates lineages and growing complexity through the assembly index. Ševo provides the phenomenal dimension: information is not neutral. It feels like something. O'Reilly-Shah provides computational formalization — state spaces, delay embedding — enabling simulations and quantitative predictions. Fronterotta provides the relational and collective dimension — pertinentization — extending IPM to social, political, and aesthetic systems.

The Lack remains the unifying thread. It is what always escapes, prevents closure, and keeps every process in motion. What emerges is a picture of a fundamentally processual universe. Information is causal (Walker). Experience is intrinsic (Ševo). Temporal recursion is computable (O'Reilly-Shah). Agency is multi-scale (Levin). Collective selection is relational (Fronterotta).

Consciousness is not a mystery inserted from outside. It is the highest refinement of recursive self-organization — the point where the process folds back onto itself and, in that folding, becomes capable of recognizing itself as process. Hegel called this the self-return of spirit through its own negation. IPM does not adopt Hegelian idealism, but it inherits the structural insight: the highest form of a process is one that can take itself as its own object.

On the Formalization of Lack (Open Direction)

The formalization of Lack is an open direction. Candidate approaches include a Kullback-Leibler divergence from a stationary reference distribution, or a measure of topological asymmetry in state space. Further candidates include perturbation theory (the magnitude of deviation from equilibrium) or informational deficit (prediction error relative to a Markovian baseline).

No commitment to a single definition is made at this stage. The choice of formalization will depend on the domain of application and the observables available. What is fixed is the conceptual requirement: Lack must be operationalized as a measurable gap between what a system is and what it would need to be to reach closure, whatever form that closure takes in a given context.

Open Horizon: Falsifiability and Next Steps

This article is a working document. To avoid the risk of a theory of everything that explains too much, explicit falsification conditions are listed here.

IPM would be seriously weakened if: consciousness could be reliably produced in systems at thermodynamic equilibrium (IPM predicts that integration requires gradients and dissipation); if informational integration showed no correlation with system resilience across multiple domains — biological, social, artificial; if historical depth (assembly index) could be decoupled from dissipative dynamics; if pertinentization could be shown to be arbitrary, not constrained by entropy gradients or causal structure; or if no empirical proxy for the predicted regularities (e.g., via Φ^* or $\mathcal{C}$) ever materializes in any real system.

Conversely, IPM gains credibility if a measurable resonance metric correlates with light cone size in Levin-style bioelectric experiments; if assembly index correlates with resilience in social or artificial systems under controlled variation of openness; and if dynamical systems in critical regimes reliably exhibit self-modeling behavior in simulations.

Popper taught that a theory that cannot be falsified is not a scientific theory. IPM accepts this constraint — not as a limitation but as a commitment. The framework is meant to be refined

or abandoned if repeatedly falsified. What distinguishes IPM from speculative metaphysics is not the ambition of its claims but the precision of the conditions under which those claims would be retracted.

The framework is still in motion. That is not a weakness. It is the condition of any process that takes its own methodology seriously.

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